Art-Style Classification Using EfficientNet-B0

Technical Report

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This technical report presents the development and evaluation of learning-based approaches for art-style classification on a severely imbalanced dataset. The work centers on EfficientNet-B0 and compares two paradigms: direct classification and metric learning. In addition, embeddings extracted from the fine-tuned EfficientNet-B0 are used to train a GradientBoostingClassifier as a downstream model. The report covers data preparation and augmentation, training methodology and hyperparameter optimization, feature-space analysis, and a comparative evaluation of model performance.

1 Introduction

Art-style classification focuses on assigning an artwork image to a predefined style category based on its visual appearance, such as color usage, composition, and subject matter. While modern convolutional neural networks perform well on large datasets, training reliable models becomes more difficult when the available data is limited and class distributions are highly imbalanced.

This project explores several learning-based approaches for art-style classification using EfficientNet-B0 as the backbone model. Three strategies are compared: direct supervised classification, metric learning with a Triplet Margin Loss, and a downstream GradientBoostingClassifier trained on embeddings extracted from the fine-tuned network. The goal is not only to compare predictive performance, but also to understand how different objectives affect the learned feature representations.

A major challenge in this dataset is the combination of few training samples and strong class imbalance. To address this, I focused on careful data preparation, controlled augmentation, and regularized fine-tuning. In addition to evaluating classification metrics, I also analyze the structure of the learned embedding space to better understand model behavior.

Main contributions

- An offline augmentation and balancing strategy designed for a small, color-sensitive dataset.
- A training setup using progressive unfreezing and Bayesian hyperparameter tuning to stabilize fine-tuning.

- A comparison between direct classification and metric learning under limited-data conditions.
- Analysis of learned embeddings using visualization, clustering, and a downstream classifier.

2 Data Preparation

Before training, the dataset was analyzed to examine its structure and identify key challenges relevant to model development.

2.1 Exploratory Data Analysis (EDA)

The dataset's structure, class distribution, image dimensions, and color characteristics were analyzed using statistical methods.

Key findings

- **Limited data:** the dataset contains few samples.
- **Severe class imbalance:** the largest class has $\sim 5\times$ more samples than the smallest. We therefore use the macro-F1 score to ensure that performance is evaluated equally across all classes, preventing dominant classes from disproportionately influencing the overall metric.
- **Color-sensitive task:** color is a key discriminative cue, so augmentation must modify color conservatively to avoid corrupting labels.

Conclusion Class imbalance is the main concern. It directly impacts the preprocessing pipeline and motivates the choice of evaluation metrics.

2.2 Data Augmentation and Preprocessing

Data split. The dataset was split into train/validation/test using a 70/15/15 ratio with stratification to preserve class proportions across splits.

Augmentation strategy To mitigate class imbalance, an offline augmentation pipeline was used to increase the number of samples in minority classes. Instead of enforcing a perfectly uniform distribution, a dynamic target count was defined for each class based on a baseline slightly larger than the majority class. A small stochastic jitter was added to these targets to avoid introducing an artificial distribution shift. Augmented samples were generated until each class reached its individual target size, producing a nearly balanced training set.

Alternative imbalance-handling methods were also explored. Uniform augmentation did not improve class balance because it produced equal numbers of augmented samples for all classes. Weighted sampling and inverse-frequency class weights were evaluated as well, but resulted in lower macro F1-scores compared to offline balancing. This suggests that increasing effective sample diversity through augmentation was more beneficial than modifying loss weights for this dataset.

2.3 Dataset finalization

All images were resized to 128×128, and colors were normalized using ImageNet mean and standard deviation. The final balanced dataset was organized into train/validation/test splits:

Split	Number of Images
Train	2,246
Validation	214
Test	214

Table 1: Dataset split used for training and evaluation.

3 Modeling Approaches

EfficientNet-B0 was selected as the backbone architecture due to its strong transfer learning performance and parameter efficiency (Tan and Le, 2019). Compared to older CNN architectures, EfficientNet achieves competitive accuracy with fewer parameters, which helps reduce overfitting risk on small datasets. EfficientNet-B0 reaches approximately 77% top-1 accuracy on ImageNet while maintaining a relatively compact model size.

Two modeling paradigms were explored: direct supervised classification and metric learning. For direct classification, the final layer of EfficientNet-B0 was replaced with a linear head predicting eight art-style classes. For metric learning, the classifier head was replaced with a linear projection head that outputs a compact embedding vector, where the embedding dimension was treated as a hyperparameter.

3.1 Direct Classification

Loss function. Cross-Entropy Loss was used for multi-class classification. Because the dataset was balanced via offline augmentation, class weights were not applied.

Optimizer. AdamW was chosen due to its decoupled weight decay, which often improves generalization during fine-tuning.

Learning-rate scheduling. Cosine annealing was used so that learning rates start higher and decrease smoothly, allowing faster early learning and more stable later fine-tuning.

Early stopping. A custom `EarlyStopper` monitored validation performance with two conditions:

- stopping when the primary validation metric plateaued,
- stopping early when the train–validation gap exceeded a threshold, indicating overfitting.

Progressive unfreezing. Training began with only the classification head unfrozen. Earlier layers were gradually unfrozen in stages to stabilize optimization.

Hyperparameter tuning. Bayesian optimization with Optuna was used to tune learning rate, batch size, weight decay, and the number of unfrozen layers. After 100 trials, the best configuration maximizing validation macro-F1 was selected.

3.2 Metric Learning

Instead of predicting class probabilities directly, this approach learns an embedding space where images from the same class are close and different classes are separated.

Loss function. Triplet Margin Loss was used with anchor, positive, and negative samples to encourage intra-class similarity and inter-class separation.

Triplet mining and sampling. Hard triplet mining selected challenging negatives within each batch, and an `MPerClassSampler` ensured multiple samples per class for effective mining.

Loss function. Triplet Margin Loss was used to learn an embedding space where samples from the same class are closer than samples from different classes. Given an anchor image x_a, a positive sample x_p from the same class, and a negative sample x_n from a different class, the objective encourages the distance between the anchor and positive to be smaller than the distance between the anchor and negative by at least a margin m:

$$\mathcal{L} = \max\left(0,\ d(f(x_a), f(x_p)) - d(f(x_a), f(x_n)) + m\right),$$

where $f(\cdot)$ denotes the embedding function and $d(\cdot, \cdot)$ is the distance metric (Euclidean distance in this work). This objective promotes compact clusters for each style while separating different styles in the embedding space.

Optimization and scheduling. AdamW and cosine annealing were used, matching the direct classification setup.

Early stopping. Validation performance was estimated using a k-NN classifier trained on embeddings. Training stopped when macro-F1 plateaued or when the train–validation gap became too large.

Hyperparameter tuning. A separate Optuna study (100 trials) tuned learning rate, batch size, weight decay, embedding size, triplet margin, and unfrozen layers. Model selection was based on validation macro-F1 computed from k-NN predictions.

4 Experimental Results

4.1 Quantitative Performance

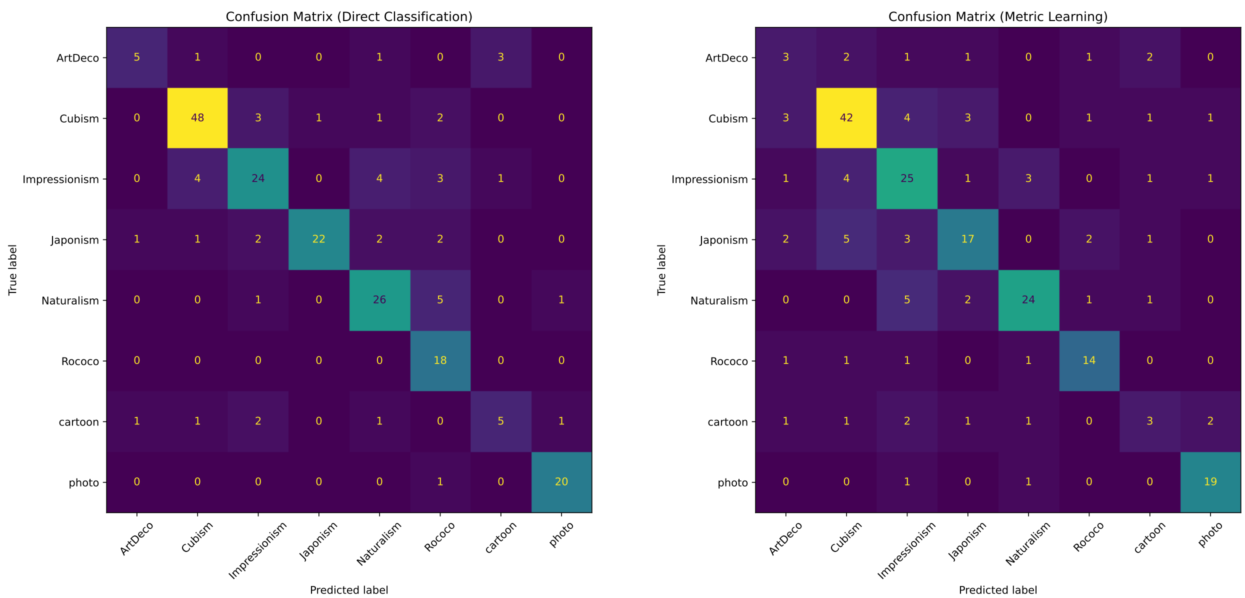
Both models were evaluated on the test set. Macro F1-score was used as the primary metric because it balances performance across classes in the presence of imbalance.

The direct classifier outperformed the metric learning approach in average by approximately $10-15$ percentage points in macro F1-score. The gap is consistent across accuracy, precision, and recall.

Per-class metrics reveal that the direct classifier is consistently stronger on most styles, with the largest improvements on minority classes such as *ArtDeco* and *cartoon*. Both approaches achieve

Metric	Direct Classification	Metric Learning
Accuracy	0.82	0.69
Macro Precision	0.85	0.63
Macro Recall	0.78	0.63
Macro F1	0.79	0.62

Table 2: Overall performance comparison between direct classification and metric learning on the test set.



(a) Direct classification

(b) Metric learning

Figure 1: Confusion matrices for direct classification and metric learning models.

strong performance on *photo* and *Cubism*, which appear visually more separable in the feature space.

Figure 1 shows confusion matrices for both approaches. The direct classifier exhibits stronger diagonal dominance, while the metric learning model produces more diffuse predictions across styles.

4.2 Training Stability and Overfitting

The direct classifier showed relatively stable training dynamics. Overfitting emerged gradually and was controlled by the custom early stopping criterion. During early training, training loss was higher than validation loss and training macro-F1 was initially lower than validation macro-F1, indicating the model did not begin in an overfit regime.

Training macro-F1 increased rapidly while validation macro-F1 improved more slowly. The early stopper triggered at epoch 3 due to an increasing train–validation gap, and the checkpoint from epoch 2 was retained.

In contrast, the metric learning model showed unstable behavior and overfit almost immediately. The train–validation gap widened quickly, causing early stopping after a single epoch across multiple runs.

4.3 Objective Comparison

The direct classifier optimizes a straightforward objective: mapping an image directly to one of eight classes. The metric learning model instead optimizes an embedding structure where distances encode class relationships. For this fixed-label classification setting with limited data, direct classification proved more reliable and data-efficient. The relatively small number of real samples likely limited the effectiveness of metric learning, which typically benefits from larger and more diverse datasets.

5 Feature Representation Analysis

After selecting the direct EfficientNet-B0 classifier as the strongest model, I analyzed the structure of its learned feature space. The goal was to verify whether the network learned meaningful style representations rather than simply memorizing labels. This was done through embedding visualization, unsupervised clustering, and downstream classification experiments.

5.1 Methodology

Feature extraction. The final classification layer was replaced with an `Identity` layer, converting the network into a feature extractor. EfficientNet-B0's penultimate layer contains 1,280 channels, so each image was represented by a 1,280-dimensional embedding vector.

Visualization with UMAP. UMAP was used to project embeddings into two dimensions for visual inspection of class structure and separability.

Unsupervised clustering with K-Means. K-Means clustering with $K = 9$ was applied to the embeddings. Eight clusters correspond to known art styles, while an additional "catch-all" cluster captures ambiguous or mixed-style samples. Clustering quality was evaluated using Adjusted Mutual Information (AMI).

5.2 Embedding Structure and Clustering Analysis

Figure 2 shows the 2D UMAP projection colored by true art style. Several styles form compact regions, while a mixed central area indicates overlapping visual characteristics. In particular, *Rococo* forms a clearly isolated cluster, suggesting strong visual consistency in that class.

Rococo likely separates well because many Rococo images in this dataset are portraits, while other styles contain more varied subjects. This suggests that the learned embeddings capture higher-level semantic cues such as subject matter and composition, not only low-level texture or brushwork.

Figure 3 shows the same projection colored by K-Means cluster assignments. Most clusters align well with true styles, forming coherent "islands," while the central overlap region feeds the additional catch-all cluster, which contains visually ambiguous samples.

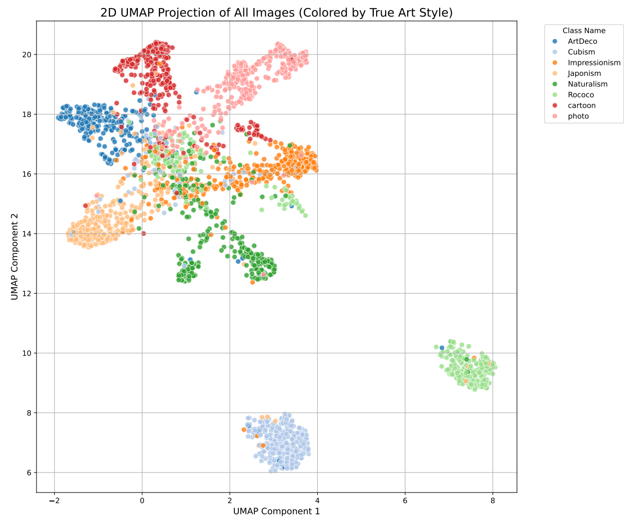


Figure 2: 2D UMAP projection of embeddings colored by true art style.

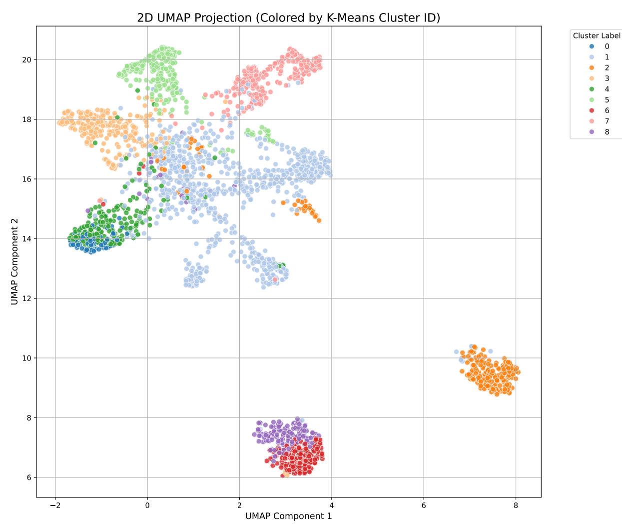


Figure 3: 2D UMAP projection colored by K-Means cluster assignment.

Overall agreement between clusters and labels was moderate (ARI ≈ 0.50), indicating that the embedding space captures meaningful structure while still reflecting blended style boundaries.

To further investigate which visual cues dominate the embedding space, one could apply controlled perturbations such as grayscale conversion (testing reliance on color), heavy blur (testing reliance on texture), or face/background masking (testing reliance on subject layout). A collapse in class separation after a perturbation would suggest that the removed cue is highly influential.

5.3 Classification on Feature Vectors (Gradient Boosting)

To test whether the embeddings contain sufficient information for classification independently of the neural network head, a `GradientBoostingClassifier` was trained on the extracted feature vectors. Hyperparameters were tuned using `RandomizedSearchCV` (50 iterations, 3-fold cross-validation) with macro-F1 as the objective. Tuned parameters included `n_estimators`, `learning_rate`, `max_depth`, and `min_samples_split`. The final model was trained on the full training embeddings and evaluated on test embeddings.

Metric	Gradient Boosting
Accuracy	0.81
Macro Precision	0.81
Macro Recall	0.76
Macro F1	0.77

Table 3: Overall performance of the embedding + Gradient Boosting model on the test set.

The Gradient Boosting model achieved an overall accuracy of 0.81 with macro F1 of 0.77. Performance was strongest on *photo*, *Cubism*, and *Naturalism*. The main remaining weakness was *cartoon*, which showed high precision but low recall, indicating that the model predicted this class conservatively.

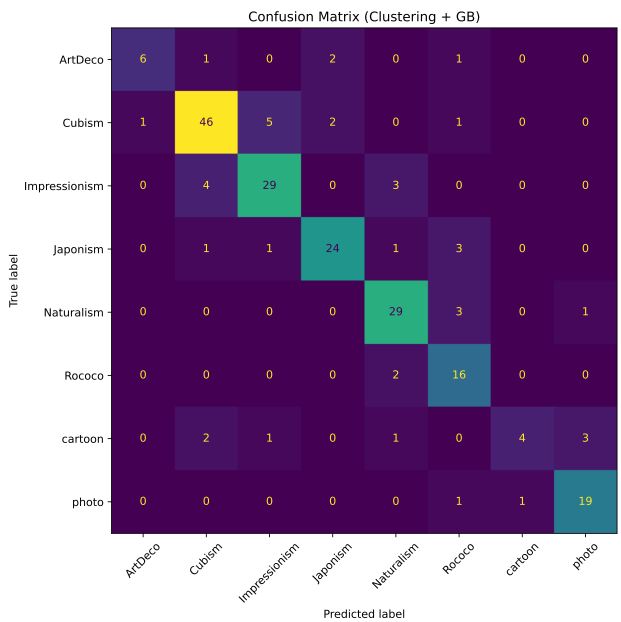


Figure 4: Confusion matrix for the embedding + Gradient Boosting pipeline.

Figure 4 shows the confusion matrix for the embedding + Gradient Boosting pipeline. Most errors occur between visually similar styles (e.g., overlap between *Cubism* and *Impressionism*), while *cartoon* is frequently confused with other styles rather than being recovered as its own class.

6 Conclusion and Future Work

Across all experiments, the main challenge was reliable prediction of underrepresented classes. Weighted sampling and inverse-frequency losses did not improve performance and often degraded stability, while offline augmentation produced more balanced learning dynamics. However, synthetic balancing cannot fully replace real data diversity, and the model may partially rely on augmentation artifacts instead of true stylistic cues.

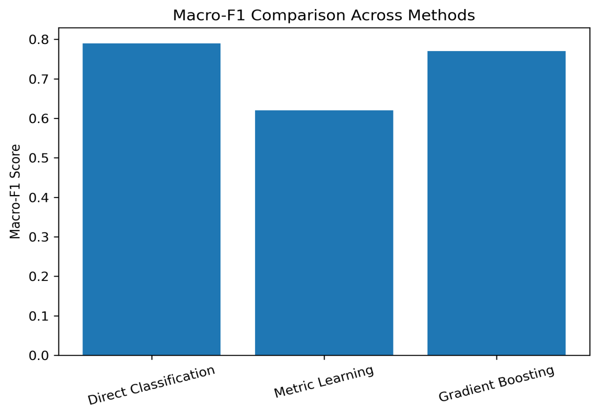


Figure 5: Macro-F1 comparision across methods.

Among the evaluated approaches, metric learning performed worst, likely because it introduces a more complex objective that is less suited to a small fixed-label classification setting. Direct fine-tuning of EfficientNet-B0 achieved the strongest results, while training a Gradient Boosting classifier on extracted embeddings produced competitive but slightly lower performance. These results suggest that end-to-end fine-tuning allows subtle feature adjustments that improve classification beyond using fixed representations alone.

Future Work

Several directions could extend this work:

- **Explainability:** apply Integrated Gradients or similar attribution methods to identify which image regions influence predictions.
- **Feature probing:** re-run embedding extraction and UMAP after grayscale, blur, or cropping perturbations to test reliance on color, texture, and subject layout.
- **More data:** improve performance by collecting additional real samples for minority classes, especially *ArtDeco* and *cartoon*.
- **Pretraining on related datasets:** warm-start the model on a larger art-style dataset (e.g., WikiArt) before fine-tuning.
- **One-vs-rest classifiers:** explore binary classifiers per class to potentially improve minority-class performance (Rifkin and Klautau, 2004).

7 Implementation Details

All experiments were implemented in PyTorch (`torch`, `torchvision`) using custom training loops with progressive unfreezing and early stopping. Image preprocessing and augmentation were built with `albumentations`, including offline balancing of minority classes. Models were optimized using AdamW with cosine learning-rate scheduling, and hyperparameters were tuned with Optuna. Metric learning used Triplet Margin Loss with hard mining and balanced sampling. Evaluation metrics were computed with `sklearn.metrics`, visualizations were generated with `matplotlib`, and embedding analysis was performed using UMAP, K-Means clustering, and a `GradientBoostingClassifier` tuned via `RandomizedSearchCV`.

References

Ryan Rifkin and Aldebaro Klautau. In defense of one-vs-all classification. *Journal of Machine Learning Research*, 5:101–141, 2004.

Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. *arXiv preprint arXiv:1905.11946*, 2019.